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Charge Inversion of Single-Stranded DNA under High-Concentration Monovalent Salts

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Cite This: *Macromolecules* 2025, 58, 12882−12892

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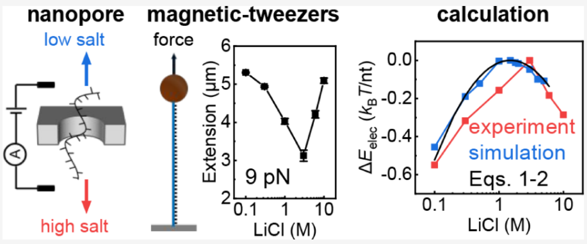


ABSTRACT: Single-stranded DNA (ssDNA) plays important roles in biology and is widely used in nanotechnology, biosensors, and drug delivery, where its functions depend critically on its physical properties. Here, we unexpectedly found that at high monovalent salt concentrations, ssDNA undergoes charge inversion, effectively switching from negatively to positively charged. This effect was consistently observed across three independent assays. Nanopore electrophoresis and dynamic light scattering experiments showed reversed electrophoretic mobility, shifting from anode- to cathode-directed migration at high salt concentrations. Single-molecule magnetic tweezers further revealed that ssDNA unexpectedly elongates and stiffens under high-salt conditions, as indicated by increased extension and persistence length. Complementary molecular dynamics simulations showed that excess cations accumulate near the phosphate backbone, reversing the net charge to approximately +0.3e per nucleotide and increasing intrachain electrostatic repulsion by ∼0.2 k_BT per nucleotide. These findings uncover a previously unrecognized electrostatic regime for ssDNA, provide new insights into ion-nucleic acid interactions, and enable programmable control of DNA behavior in high-salt environments, with implications for nanodevice engineering and biomolecular regulation.

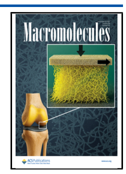
■ INTRODUCTION

Single-stranded DNA (ssDNA) plays an essential role in both biological systems and DNA nanotechnology. It naturally arises as an intermediate in key cellular processes, including replication, transcription, and DNA repair, and is a major component of telomeres and many viruses.[1−4] For example, during the repair of double-stranded breaks, homologous recombination is initiated through DNA end resection, generating ssDNA overhangs at the break sites.[5] In synthetic systems, ssDNA is widely used to construct programmable nanostructures such as DNA origami and bricks,[6,7] where precise control over sequence and length enables the assembly of complex architectures. Understanding the conformation and elasticity of ssDNA is critical for elucidating its function in both biological and engineered contexts.

Owing to its highly negative charge and intrinsic flexibility, ssDNA adopts conformations that are highly sensitive to environmental conditions such as ionic strength, temperature, and applied force.[8,9] The mechanics of ssDNA are commonly described using polymer physics models such as the Freely Jointed Chain (FJC) and Wormlike Chain (WLC) models, which characterize chain flexibility through the Kuhn length and persistence length, respectively.[10−12] Various experimental techniques have been developed to probe ssDNA conformations. Single-molecule Förster resonance energy transfer (smFRET) and ensemble techniques like small-angle X-ray scattering (SAXS) have revealed how monovalent salt modulates the compactness and persistence length of short ssDNA.[13−17] Notably, studies by Murphy et al. and Chen et al. showed a monotonic decrease in persistence length with increasing salt concentration for short poly dT ssDNA using smFRET and SAXS, respectively.[13,14] More recently, Roth et al. developed a DNA-origami-based approach that enabled direct measurements of ssDNA persistence length across a broader range of lengths.[18]

Beyond the characterization of ssDNA without external forces, single-molecule manipulation experiments such as magnetic tweezers (MT), optical tweezers (OT), and atomic force microscopy (AFM) provide direct insights into the elastic response of ssDNA under tension.[16,19−23] These studies have revealed a nonlinear force−extension relationship: a power-law regime at low forces and a logarithmic response at higher forces, consistent with WLC predictions.[16,24] A key finding by Saleh et al. was that the effective Kuhn length of ssDNA scales linearly with the Debye length, deviating from the classical Odijk−Skolnick−Fixman (OSF) model. They proposed that internal electrostatic tension plays a significant role in the intermediate force regime.[19] This idea was extended in later work, which identified a universal electrostatic stiffening exponent of ∼0.66 for ssDNA in monovalent salts.[20]

Received: August 15, 2025
Revised: October 17, 2025
Accepted: November 13, 2025
Published: November 18, 2025



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Complementary OT studies by Ritort and colleagues further highlighted how base stacking and secondary structure formation influence ssDNA elasticity in both monovalent and divalent salt environments.[21,22] While these findings offer a comprehensive picture of ssDNA elasticity at low to moderate ionic strengths (up to ∼3 M), the behavior of ssDNA under extreme salt conditions remains unclear.

In this work, we provide conclusive evidence that monovalent salts at high concentrations can induce charge inversion and lead to unexpected nonmonotonic behaviors in ssDNA conformation and electrophoretic mobility by employing three complementary experimental approaches. First, our nanopore translocation measurements revealed a striking reversal in the direction of ssDNA migration: above 4 M LiCl, translocation shifted from anode-directed to cathode-directed, indicating the charge inversion of ssDNA. Second, the dynamic light scattering (DLS) measurements of electrophoretic mobility of ssDNA confirmed this transition across multiple monovalent salts (LiCl, NaCl, RbCl, CsCl, and KAc), suggesting a general mechanism instead of ion-specific effects. Third, single-molecule magnetic tweezers experiments showed that both end-to-end extension and persistence length of ssDNA unexpectedly increased at high monovalent salt concentrations. These results were corroborated by our all-atom molecular dynamics (MD) simulations, which demonstrated that excess cation accumulation around the phosphate groups reverses the net charge of the molecule. Our experimental and simulation results further estimate that charge inversion at high salt concentrations enhances the electrostatic repulsion between adjacent nucleotides by ∼0.2 kBT. Additionally, we conducted temperature-controlled MT experiments to quantify the temperature dependence of force−extension curves, which allowed us to determine the change in conformational entropy as a function of salt concentration.

Together, our results establish that charge inversion is a universal feature of ssDNA in highly concentrated monovalent salt solutions. This finding challenges prevailing views in polyelectrolyte physics and reveals a previously unrecognized regime of ssDNA behavior.[25,26] These insights advance the fundamental understanding of nucleic acid electrostatics and have significant implications for DNA-based nanotechnology, biomolecular engineering, and biological systems operating under extreme ionic conditions.

■ MATERIALS AND METHODS

Nanopore Measurements. We studied the ionic current changes during the translocation of ssDNA through the nanopore using the purchased M13mp18 single-stranded DNA plasmid (Takara Bio Inc.) at a concentration of 800 pM. All nanopore translocation measurements were performed in a buffer containing 10 mM Tris (pH 8.0) with different concentrations of LiCl. DNA characterization was performed using glass nanopores fabricated with a laser puller (P2000/G, Sutter Instruments). The instrument parameters were set as follows: HEAT 750, FIL 3, VEL 25, DEL 145, and PUL 200, resulting in nanopores with a mean diameter of ∼12 nm. Ag/AgCl electrodes were used as the electrochemical sensors. Specifically, the transend electrode was inserted into a capillary tube, while the cis-end electrode was positioned in the liquid reservoir chamber. DNA samples were introduced into the reservoir chamber, and upon their translocation through the nanopores, characteristic blockade current signals were recorded. Ionic currents were recorded and analyzed using Clampfit 10.6 software (Molecular Devices) and custom-written Matlab software. All ionic current measurements were conducted using an Axopatch 200B amplifier (Molecular Devices) in voltage-clamp mode with a low-pass Bessel filter set to 10 kHz, followed by digitization with a DigiData 1550B digitizer (Molecular Devices) at 250 kHz. The entire instrument setup was placed inside a double Faraday cage enclosure. All experiments were performed at room temperature.

Dynamic Light Scattering Experiments. We performed electrophoresis experiments using purchased M13mp18 (Takara Bio Inc.) and ΦX174 (NEB) single-stranded DNA plasmids at a concentration of 2 nM. The experiments were conducted at 25 °C in 10 mM Tris-HCl (pH 8.0) buffer with a total volume of 700 μL, containing different concentrations of ions. The electrophoretic mobility of each mixture was measured at 298.15 K by using a Nano-ZS90 Zetasizer (Malvern). Positive values of μ correspond to DNA migrating toward the cathode.

Buffer Solutions. We prepared a near-saturation concentration of the salt mother liquor, which underwent filtration and autoclaving. Before each experiment, we diluted the mother liquor to different salt concentrations using deionized water. A consistent pH of 8.0 was maintained by using 10 mM Tris-HCl.

Magnetic Tweezers Experiments. Our in-house developed MT setup utilized an inverted microscope (IX73, Olympus) featuring a 60× oil immersion objective lens (Olympus), upon which a flow cell was mounted. Precise vertical positioning of the objective was achieved via a Piezo objective scanner (P-721, Physik Instrumente). A pair of vertically aligned neodymium magnets (5 mm cubes), separated by a 1 mm gap, was used to attract the paramagnetic beads and apply consistent forces to the DNA molecules. The linear motor (L-509, Physik Instrumente) controlled the stretching force on DNA by adjusting the vertical position of the magnets. A red collimated LED light source illuminated the observation area through the gap between the magnets. Bead tracking was conducted using a CCD camera operating at 100 frames per second (MER2-230-168U3M, Daheng Imaging), and bead positions in three dimensions were calculated by using a previously established algorithm.

We designed 5′ thiol- and 3′ DBCO-labeled DNA capable of withstanding experiments for hours under high ion concentrations (Figure S1). We coated the APTES-functionalized slide with sulfo-SMCC (Thermo Fisher Scientific) inside the flow cell. The thiol end of the long dsDNA was anchored to the slide, and the surfaces were passivated for 2 h using a solution of 2% bovine serum albumin (Sigma-Aldrich) and 0.1% 2-mercaptoethanol (Sigma-Aldrich) in phosphate-buffered saline. Meanwhile, we used NHS-PEG-azide to modify the amino-functionalized paramagnetic microbead (Dynabeads, M-270), which was attached to the DBCO end of a short dsDNA. Finally, the short and long dsDNA fragments were ligated using T4 DNA ligase in the flow cell (Thermo Fisher Scientific). We peeled off the untethered ssDNA strand in the dsDNA at 40 mM NaOH and left the tethered ssDNA stretched at ∼40 pN. Detailed protocols of the ssDNA sample preparation and the sequences of oligos used are available in the supplementary text.

All-Atom Molecular Dynamics Simulations. The 20-nt ssDNA was built using 3DNA with the sequence CGACTCTACGGCATCTGCGC.[27,28] The ssDNA molecule was

https://doi.org/10.1021/acs.macromol.5c02249
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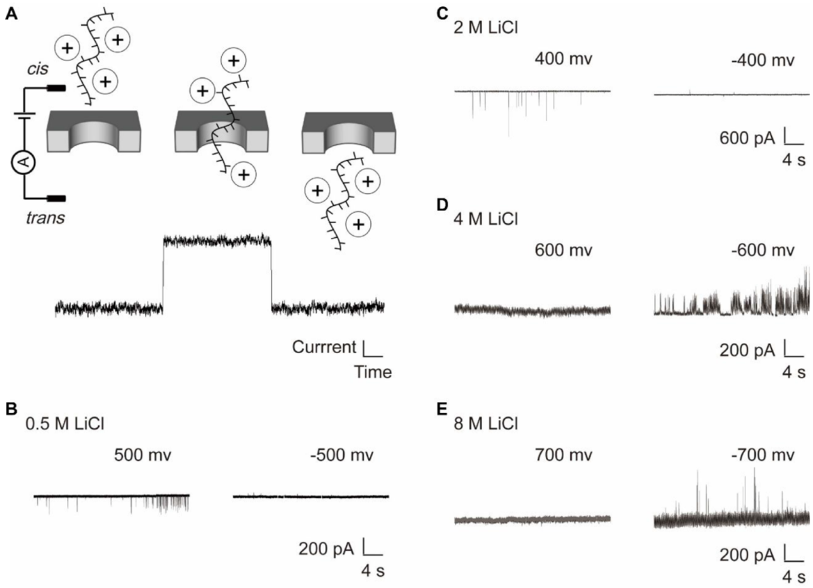

Figure 1. Nanopore measurements show the charge inversion of ssDNA. (A) Schematic of the nanopore experiment. Current blockades are produced when ssDNA molecules enter the pore. (B-E) Current−time traces of 7.2k-nt ssDNA in 10 mM Tris-HCl (pH 8.0) at varying LiCl concentrations.

initially placed in a rectangular box ($8 \times 8 \times 10$ nm^3) containing TIP3P water molecules. Then, cations and anions were added to neutralize the net charge of the ssDNA. For each simulation system with different salt concentrations, different numbers of cation and anion pairs were added to the simulation box corresponding to the bulk concentrations (Table S1). For each system, energy minimization was first employed to avoid unfavorable strains or contacts. Next, the system was equilibrated in the canonical ensemble (NVT) at 295 K for 500 ps using a V-rescale thermostat and in the isothermal−isobaric ensemble (NPT) at 1 bar for 10 ns using Parrinello−Rahman barostat.[29,30,39] Finally, a production run of 600 ns was performed for each system. During production MD, a constant stretching force was applied to one end of the ssDNA, while the other end was fixed with a strong harmonic constraint.

All-atom MD simulations were performed with the GROMACS 2021 software package and the OL15 force field.[31,32] The steepest descent algorithm was used for energy minimization. An integration step of 2 fs was used in conjunction with the leapfrog algorithm. The LINCS algorithm was used to impose constraints on all bonds connecting to hydrogen atoms. A 1.0 nm cutoff was applied for van der Waals interactions and short-range electrostatic interactions. The long-range electrostatic interactions were calculated using the particle mesh Ewald (PME) method.[33] The configuration was saved every 10 ps, giving 60,000 snapshots in total for analysis.

Coarse-Grained Molecular Dynamics Simulations. We also performed coarse-grained MD simulations of a 100-nt ssDNA molecule with different salt concentrations (0.1, 0.3, and 1.0 M) using the oxDNA2 model.[34] Initially, we constrained the movements of 5-nt at the bottom end of the ssDNA and stretched 5-nt of the other end in the z-direction. Different tension forces (0−20 pN) were applied to the ssDNA. This setup mimics the behavior of ssDNA in the MT experiments. After obtaining the equilibrium configuration from Monte Carlo simulations of 10,000 steps, we performed the MD simulations of 10^9 steps, which give a simulation time of 5×10^6 or a real time of 15 μs. We used the Anderson-like thermostat with a time step of 0.005 in reduced time units at 295 K.[35] The conversion between oxDNA reduced units and SI units is given in Table S2. Because there are no explicit ion particles in oxDNA simulations and the electrostatic interactions are treated using the Debye−Hückel model, the simulations were conducted in salt concentrations below 1 M.

RESULTS

Electrophoretic Direction Reverses with Increasing Concentrations of Monovalent Salts. To investigate the electrostatic behavior of ssDNA under varying salt conditions, we first examined its electrophoretic direction by using solid-state nanopore experiments. We conducted the experiments using M13mp18, a 7.2k-nt circular ssDNA isolated from the M13 bacteriophage (Takara Bio Inc.). The measurements were carried out using ∼12 nm inner diameter glass nanopores, with Ag/AgCl electrodes serving as the electrochemical interfaces. The transside electrode was positioned within a capillary tube, while the cis-side electrode was placed in a fluid reservoir chamber.[36]

The ionic current was continuously recorded through the voltage-biased nanopore. In the absence of ssDNA, a stable baseline current was recorded under both positive and negative voltages. Once ssDNA was introduced into the reservoir, translocation events were detected as transient blockades in the ionic current. The applied voltage determined the direction of DNA movement, reflecting its net charge: negatively charged ssDNA was drawn through the nanopore under positive bias, whereas positively charged ssDNA migrated under negative bias, producing characteristic blockade signals (Figure 1A). At low salt concentrations (0.5 and 2 M LiCl), we observed robust current blockades under positive voltage, while no events were detected under negative voltage (Figure 1B−C), confirming the expected negative charge of ssDNA under these conditions.[37,38] Strikingly, when the LiCl concentration was increased to 4 and 8 M, the translocation events occurred predominantly under negative voltage, with few or no events

https://doi.org/10.1021/acs.macromol.5c02249
Macromolecules 2025, 58, 12882−12892

observed under positive bias (Figure 1D–E). This reversal of translocation polarity indicates a net positive charge on the ssDNA, demonstrating charge inversion induced by high concentrations of monovalent salt. Notably, the effective charge of ssDNA is strongly screened at 4 M LiCl, so electrophoretic driving is weak and translocation is slower. Molecules more frequently undergo brief stalling or partial entry-exit near the pore mouth, producing longer or multisegment blockades; as a result, the trace appears more populated, yet the current often fails to return to baseline, indicating incomplete translocations.[39,40] By contrast, at 8 M, events are typically single-pass for each molecule.

No charge inversion was observed in KCl solutions of up to 4 M (Figure S4), which is close to the solubility limit of KCl, whereas weak inversion was already detectable in 4 M LiCl (Figure 1D). This finding is consistent with the weaker ion-DNA interaction of K^+ compared to Li^+, as supported by previous molecular dynamics and nanopore studies.[39,41]

To corroborate our findings on ssDNA charge inversion, we performed dynamic light scattering (DLS) measurements to quantify the electrophoretic mobility (μ) of 7.2k-nt M13mp18 ssDNA under various monovalent salts. We measured μ for different monovalent salts (LiCl, NaCl, KCl, RbCl, and CsCl) to exclude specific interactions between LiCl and ssDNA. The sign of μ reflects the direction of electrophoretic motion and thus the net charge of the molecule, where a reversed sign of μ indicates charge inversion of DNA.[42] Figure 2A shows that μ

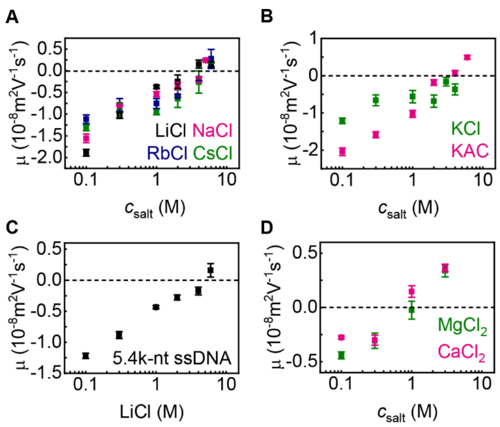

Figure 2. The electrophoretic mobility of ssDNA indicates the charge inversion of ssDNA. The error bars represent the standard errors from more than three independent experiments in 10 mM Tris-HCl (pH 8.0) buffer. (A-B) Electrophoretic mobility (μ) of 7.2k-nt ssDNA at different concentrations of monovalent salts. (C) μ of 5.4k-nt ssDNA at different concentrations of LiCl. (D) μ of 7.2k-nt ssDNA at various concentrations of divalent salts ($MgCl_2$, $CaCl_2$).

was consistently negative up to ∼3 M for all tested monovalent salts, indicating negatively charged ssDNA. Beyond this threshold concentration, μ was reversed to positive values, corresponding to positively charged ssDNA. Our measurements showed that the charge inversion of ssDNA occurred at high concentrations of all monovalent salts (LiCl, NaCl, RbCl, and CsCl) except KCl. The lack of mobility reversal in KCl is likely due to its limited solubility in water. To overcome this, we carried out electrophoresis experiments using KAc, which enabled measurements at higher concentrations and manifested charge inversion of ssDNA above 3 M (Figure 2B), consistent with the results obtained using other monovalent salts. Due to the confinement effect in the nanopore, effects like electric field strength, ion concentration, and electroosmotic flow (EOF) are significantly different from those in the bulk solution. This could lead to differences in the quantitative results compared to those characterized in the bulk solution. Therefore, we used nanopore results primarily as qualitative indicators of the net charge sign of ssDNA.

To assess whether ssDNA charge inversion depends on the sequence or length, we repeated the DLS experiments using the 5.4k-nt circular ssDNA, which is isolated from bacteriophage ΦX174 (NEB). Similar to the 7.2k-nt ssDNA, the mobility reversed from negative to positive at ≥4 M LiCl (Figure 2C). This result supports the idea that ssDNA charge inversion is a general phenomenon under high concentrations of monovalent salts rather than sequence-specific. The subtle differences in the critical concentrations of charge inversion might be caused by ion-specific effects. Furthermore, we extended our analysis to divalent salts ($MgCl_2$ and $CaCl_2$), where electrophoresis experiments revealed a more pronounced reversal in ssDNA mobility: from approximately ∼−0.5 in 0.1 M to ∼+0.5 in 3 M (Figure 2D). The electrophoretic mobility is governed by two factors: both the effective charge and the hydrodynamic radius of ssDNA. To separate these two factors, we attempted to measure the ssDNA hydrodynamic radius using DLS, but the DLS experimental results exhibited significant fluctuations, possibly due to the intrinsic conformational heterogeneity of ssDNA. This variability prevented us from establishing a precise dependence of the ssDNA hydrodynamic radius on salt concentration. Although these two factors cannot be quantitatively separated from the nanopore and DLS data, their trends are consistent with the increase of effective charge predicted by eq 1 and the modest expansion of chain dimensions from MD simulations at high salt. Taken together, our nanopore and electrophoretic mobility measurements consistently demonstrate that ssDNA undergoes charge inversion at high monovalent salt concentrations. This behavior is robust across multiple monovalent salts and ssDNA types, revealing a universal electrostatic phenomenon that challenges traditional models of polyelectrolyte behavior.

The Extension of ssDNA Reverses with Increasing Monovalent Salts in MT Experiments. To further investigate the structural response of ssDNA under high salt conditions, we conducted single-molecule MT experiments to measure the force–extension curves (FECs) of ssDNA in various monovalent salts, as illustrated in Figure 3A. We designed a 5′-thiol- and 3′-DBCO-labeled 13.6k-nt ssDNA for the experiment. The 5′-thiol group was conjugated to a maleimide-modified glass surface, while the 3′-DBCO was linked to an azide-functionalized paramagnetic microbead (Dynal bead, M-270), forming a stable, doubly covalently tethered ssDNA construct suitable for extended measurements under high salt conditions.

Figure 3B presents the representative FECs in different LiCl concentrations, where the extension of ssDNA (x) decreases monotonically with decreasing force f, consistent with entropic elasticity. The FECs of other monovalent salts are shown in Figure S5. At fixed forces, the extension of ssDNA initially decreases as salt concentration (c_{salt}) increases, up to a critical threshold concentration of 3 M (c^*_{salt}). This trend is attributed to classical electrostatic screening, which reduces repulsive interactions along the charged DNA backbone. However,

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Macromolecules 2025, 58, 12882−12892

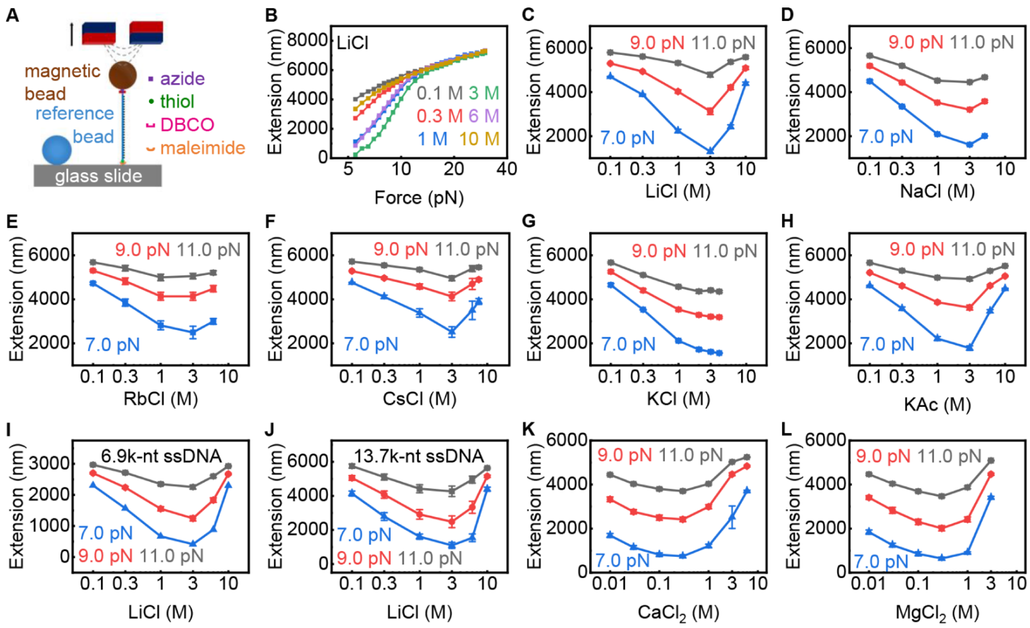

Figure 3. MT experiments indicate the charge inversion of ssDNA. Extension offsets were calibrated by aligning the same ssDNA tether (3 molecules) at 60 pN across different mono- and divalent salts, respectively. The error bars represent the standard errors from three independent experiments. (A) Schematic illustration of the MT experiment setup used for ssDNA manipulation. (B) Force−extension curves of ssDNA measured in varying LiCl concentrations at 23 °C. (C−H) The extension of 13.6k-nt ssDNA as a function of c_{salt} for different monovalent salts under constant stretching forces (7, 9, and 11 pN). (I−J) The extension of (I) 6.9k-nt ssDNA and (J) 13.7k-nt ssDNA as a function of c_{salt} for LiCl under constant stretching forces (9 and 11 pN). (K−L) The extension of 13.6k-nt ssDNA as a function of c_{salt} for (K) $CaCl_2$ and (L) $MgCl_2$.

beyond c^*_{salt}, a reversal is observed: the ssDNA extension increases with further increases in c_{salt}. This nonmonotonic behavior suggests that charge inversion occurs at high salt concentrations, leading to a net overcompensation of negative charges by counterions, thereby reintroducing effective repulsive forces between monomers.

To systematically quantify this phenomenon, we extracted ssDNA extension as a function of c_{salt} under three constant forces (7, 9, and 11 pN), as shown in Figure 3C−L. The formation of secondary structure was eliminated by calculating the fraction of unpaired ssDNA bases that do not form secondary structure as a function of force (see supplementary text of SI). As shown in Figure S2, the fraction increases significantly at low forces (≤7 pN) and becomes ∼1 at forces ≥9 pN for almost all c_{salt}. This result suggests that secondary structure formations are not taken into account in the variation of ssDNA extension with c_{salt} at larger forces (9 and 11 pN). We did not use the high-force regimes because electrostatic contributions become negligible at forces ≥20 pN. For all monovalent salts, the extension of ssDNA decreased with c_{salt} up to $c^*_{salt} \approx 3$ M, followed by an increase in extension at higher c_{salt}. An exception was observed for RbCl, which showed a slightly lower c^*_{salt}, possibly indicating an earlier onset of charge inversion. Notably, KCl did not show a significant extension increase at high c_{salt}, likely due to its limited solubility. When we substituted KCl with KAc, which allows higher concentrations, the extension reversal appeared beyond 3 M (Figure 3H), consistent with those of other monovalent salts.

To rule out sequence- or length-specific effects, we repeated the MT experiments using 6.9k-nt and 13. 7k-nt ssDNA constructs in LiCl. Both exhibited the same biphasic response: extension decreased below c^*_{salt} and increased beyond it (Figures 3I−J and S7). Similar trends were also observed in $MgCl_2$ and $CaCl_2$ solutions, though with lower charge inversion thresholds of $c^*_{salt} \approx 0.3$ M (Figures 3K−L and S8), in agreement with our electrophoresis results for divalent salts.

In addition to the extension, we quantified ssDNA stiffness by fitting each FEC to the Marko−Siggia formula of the WLC model: $f = \frac{k_B T}{L_p}\left[\frac{1}{4(1 - x/L_c)^2} - \frac{1}{4} + \frac{x}{L_c}\right]$. Here, k_B is the Boltzmann constant, T is the temperature, L_p is the ssDNA persistence length, and L_c is the ssDNA contour length. The fits to FECs are shown in Figure S9, where the WLC model works well over the salt concentrations covered here. The contour length does not change much with c_{salt} (Figure S10). As shown in Figure 4, the L_p first decreases with the increasing c_{salt} for all monovalent salts, consistent with previous studies.[14,15] However, beyond $c^*_{salt} \approx 3$ M, the L_p begins to rise with increasing c_{salt}. Specifically, L_p increases from ∼0.62 nm to ∼0.72 nm when c_{salt} of LiCl increases from 3 to 10 M.

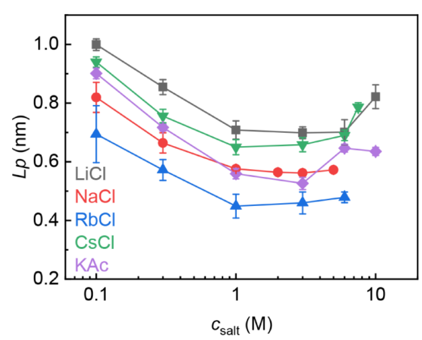

Figure 4. Persistence length of ssDNA at different monovalent salt concentrations. The error bars represent the standard errors from four independent experiments.

The slower L_p increase in NaCl compared to other salts might be due to weaker charge inversion in this salt environment.

The reversal of ssDNA L_p reflects the transition in electrostatic interactions at very high concentrations of monovalent salts. According to polyelectrolyte theories,[43−45] the L_p is a sum of the intrinsic persistence length (L_p^0) and the electrostatic persistence length (L_p^{elec}): $L_p = L_p^0 + L_p^{elec}$. L_p^0 is independent of salt concentrations, whereas L_p^{elec} arises from electrostatic repulsion between like charges within the polyelectrolyte.[43,44] At low to moderate c_{salt}, the electrostatic screening of counterions reduces L_p^{elec}, leading to lower chain stiffness. At high c_{salt}, however, charge inversion introduces net positive charges, reinstating electrostatic repulsion and increasing L_p^{elec}. Our findings refresh previous studies on the bending stiffness of ssDNA and dsDNA, whose L_p decreases monotonically with c_{salt}.[14,46]

To further quantify the salt-dependent electrostatic contributions to ssDNA mechanics, we calculated the electrostatic energy difference, ΔE_{elec}, across the examined c_{salt} by integrating force over matched extension ranges in FECs: $\Delta E_{elec} = \int_{x_1}^{x_2} f_{c_{salt}}\, dx - \int_{x_1}^{x_2} f_{c_{salt}^*}\, dx$. The salt-induced electrostatic change is the difference between the integration over the same extension range (500−6000 nm) in FECs (Figure S11). By setting $\Delta E_{elec} = 0$ at the c_{salt}^*, we assessed the relative weakening or strengthening of electrostatic interactions. Figure 6E−F presents the ΔE_{elec} as a function of c_{salt} for LiCl and NaCl, respectively. In LiCl, the electrostatic interactions are weakened by ∼0.6 k_BT/nt as c_{salt} increases from 0.1 to 3 M, followed by a strengthening of ∼0.3 k_BT/nt from 3 to 10 M. For NaCl, the electrostatic interactions are weakened by ∼0.5 k_BT/nt from 0.1 M to c_{salt}^* of 2 M, followed by a strengthening of 0.1 k_BT/nt from c_{salt}^* to 5 M. The difference in magnitudes before and after c_{salt}^* might arise from spatial constraints on counterion accumulation and differences in ion-specific hydration or distribution patterns. These results confirm that charge inversion dramatically alters the balance of electrostatic interactions in ssDNA, and that the effect varies by salt species.

MD Simulations Reveal Microscopic Origins of ssDNA Charge Inversion. While the electrophoretic and MT experiments clearly demonstrate the charge inversion of ssDNA at high monovalent salt concentrations, the atom-scale conformational and electrostatic changes associated with this phenomenon remain poorly understood. To address this, we performed all-atom MD simulations of ssDNA at varying concentrations of monovalent salts.

We first analyzed the extension of ssDNA as a function of c_{salt} from simulations (Figure 5A−B). Consistent with MT results, the extension of ssDNA in LiCl initially decreased from 0.35 to 0.23 nm as c_{salt} increased from 0.5 to 2 M. However, a further increase from 2 to 5 M led to an increase in extension from 0.23 to 0.34 nm. This trend is similar to that observed in the MT experiments. The extension reversal beyond $c_{salt}^* \approx 2$ M indicates charge inversion of ssDNA. Minor quantitative discrepancies between simulation and experiment, such as a lower c_{salt}^* and shorter absolute extension, should be caused by overestimation of interactions between Li^+ and DNA in the force field. Simulations with NaCl also reproduced the biphasic trend but exhibited a slightly different increase in extension at

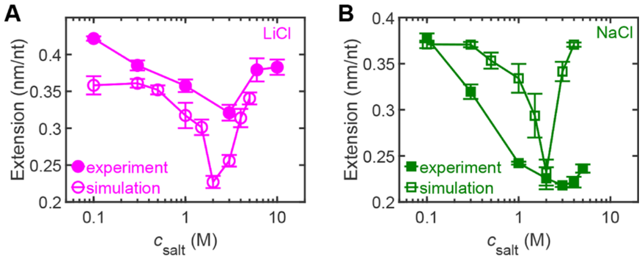

Figure 5. MD simulations reproduce reversals of ssDNA extension at very high concentrations of monovalent salts. (A−B) Comparison of the extension between MT experiments and simulations in (A) LiCl and (B) NaCl.

high salt, likely due to the shorter length of ssDNA in simulations.

Next, we calculated the radius of gyration (R_g) of the ssDNA across the same salt range (Figure S12A). At low c_{salt}, R_g decreased from 2.1 to 1.5 nm as LiCl concentrations increased from 0.1 to 3 M, or from 2.1 to 1.8 nm as NaCl concentration increased from 0.1 to 2 M, indicating ssDNA compaction due to stronger electrostatic screening. Beyond these thresholds, R_g was reversely increased to 2.0 nm as LiCl concentration increased to 5 M or to 2.1 nm as NaCl concentration reached 4 M, signifying re-expansion of ssDNA consistent with reintroduced electrostatic repulsion following charge inversion. From the R_g values, we estimated the L_p using the WLC model:[47,48] $R_g^2 = L_c L_p \left\{ \frac{1}{3} - \frac{L_p}{L_c} + \frac{2L_p^2}{L_c^2} - \frac{2L_p^3}{L_c^3}[1 - e^{-L_c/L_p}] \right\}$. As shown in Figure S12B, the L_p increased with c_{salt} beyond the charge inversion c_{salt}^*, supporting experimental observations of ssDNA stiffening at high salt.

To probe charge inversion at the molecular level, we computed the total local charge (q_{all}), the net charge within a distance (d) from the ssDNA surface. q_{all} includes the charges of DNA atoms, cations, and anions and is normalized per nucleotide. As shown in Figure 6A, at LiCl $c_{salt} \leq 1$ M, q_{all} remained negative for $d \leq 4.0$ nm, consistent with an overall negatively charged ssDNA. At higher LiCl c_{salt}, q_{all} became positive near the ssDNA surface ($0.1 \leq d \leq 1.0$ nm), indicating overcompensation by cations. The maximum local charge (q_{all}^{max}) occurred consistently at $d \approx 0.45$ nm and increased monotonically with c_{salt} (Figure S13A). A similar trend was observed in NaCl, with q_{all}^{max} localized at $d \approx 0.45$ nm after $c_{salt}^* \approx 2$ M (Figures 6B and S13B). To elucidate the mechanism, we quantified the average number of cations bound directly to the phosphate groups on ssDNA (N_{cation}), as illustrated in Figure 6C. Below the charge inversion threshold c_{salt}^*, $N_{cation} < 1$, consistent with partial electrostatic screening (Figure 6D). At c_{salt}^*, $N_{cation} \approx 1$. And $N_{cation} > 1$ above c_{salt}^*, leading to charge inversion of ssDNA. The strong binding affinity of monovalent cations to the phosphate backbone thus appears to drive the inversion of net charge. At high c_{salt}, the radial distribution function (RDF) of cations and anions around phosphates of ssDNA exhibits oscillations (Figure S14), reflecting alternating layers of ion accumulation and depletion near the phosphate backbone. Such an oscillatory feature arises from the competition between electrostatic correlation and excluded volume effects in concentrated electrolytes, consistent with theoretical predictions.[49]

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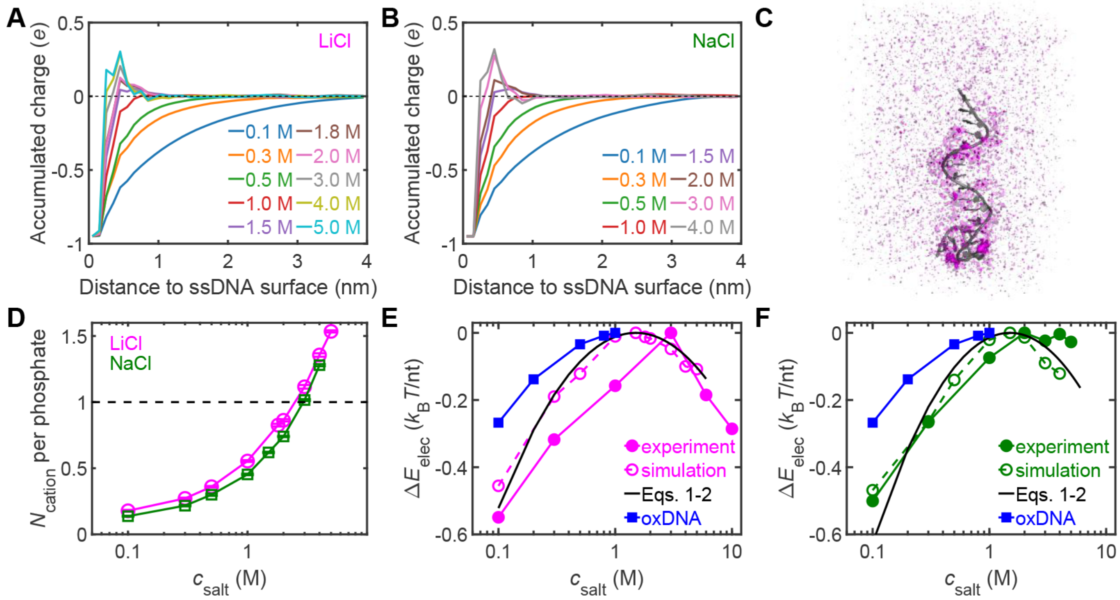

Figure 6. Charge inversion destabilizes ssDNA by weakening the electrostatics. (A–B) The accumulated charge of DNA atoms and ions averaged over 20 nucleotides in (A) LiCl and (B) NaCl. (C) Illustration of cation distribution around ssDNA. (D) The effects of c_{salt} on the number of cations around ssDNA phosphates. (E–F) The electrostatic potential as a function of c_{salt} for (E) LiCl and (F) NaCl.

We further defined the total effective charge of ssDNA, q_{DNA}, roughly equaling q_{all} at $d \approx 0.45$ nm and representing the dominant contribution to electrostatic interactions. Below c^*_{salt}, q_{DNA} becomes less negative with increasing c_{salt} due to electrostatic screening. Above c^*_{salt}, q_{DNA} becomes positive, indicating charge inversion. The correlation between q_{DNA} and c_{salt} fits well to the logarithmic equation:

$$q_{DNA} = a\ln(c_{salt}/c^*_{salt}) \tag{1}$$

with a = 0.234 for LiCl and a = 0.226 for NaCl. See more about eq 1 in the Supplementary text. We thus estimated the electrostatics of ssDNA with c_{salt} using q_{DNA}:

$$\Delta E_{elec}/k_BT = \frac{(q_{DNA}e)^2}{4\pi\varepsilon_0\varepsilon_r l} \tag{2}$$

where e is the elementary charge, ε_0 is the vacuum permittivity, ε_r is the dielectric constant of water l = 0.6 nm is ssDNA contour length per nucleotide. Note that we do not use the dielectric constant of DNA because of the narrow cross-section for ssDNA. The predictions from eqs 1-2 were in good agreement with the experimental and simulation results, as shown in Figure 6E–F.

Additionally, we performed coarse-grained oxDNA2 simulations of 100-nt ssDNA to examine salt-dependent elasticity under different stretching forces below 1 M salt (Figure S15). The oxDNA2 model is efficient at reproducing the expected electrostatic screening at low salt but failed to capture charge inversion effects beyond c^*_{salt} due to its reliance on linear screening using Debye–Hückel theory. The electrostatic potential from oxDNA2 is shown by the blue line in Figure 6E–F, which agrees with our experimental and simulation results for $c_{salt} \leq 1$ M.

Together, our simulations provide atom-level evidence of ssDNA charge inversion at high monovalent salt concentrations. The conformational expansion, mechanical stiffening, and local charge inversion observed in simulations directly corroborate experimental findings, offering a unified electrostatic framework to explain the anomalous behavior of ssDNA under extreme ionic conditions.

Charge Inversion Reduces ssDNA Conformational Entropy. We also measured FECs of ssDNA at different temperatures (30, 37, 44, and 52 °C) with the same LiCl concentrations, as shown in Figure 3B. The FEC at each temperature is similar within the high-force regime but deviates in the force regime where electrostatic interactions dominate. As shown in Figure 7A, the extension changes with c_{salt} of LiCl

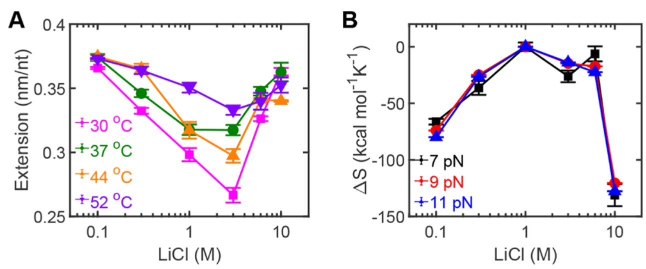

Figure 7. Temperature-controlled charge inversion of ssDNA. (A) The extension of 13.6k-nt ssDNA as a function of c_{salt} for different temperatures in LiCl under 9 pN. (B) The conformational entropy changes as a function of c_{salt} for LiCl under different stretching forces of 7, 9, and 11 pN.

are similar for different temperatures, with c^*_{salt} = 3 M for each curve. However, the curves at high temperatures show less significant change between that of c^*_{salt} and other c_{salt}. This indicates that increasing the temperature can elongate ssDNA extension and weaken the effect of electrostatic screening.

The behaviors of ssDNA are well described by the WLC model, where external forces must be applied to extend the end-to-end distance. The stretched ssDNA is entropically unfavorable and acts against the entropic force; thus, the work done by the stretching force leads to a reduction in conformational entropy. Subject to the increasing c_{salt}, the

https://doi.org/10.1021/acs.macromol.5c02249
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equilibrium extension of ssDNA between the entropic force and external stretching force is decreased due to the electrostatic screening effect. We calculated the entropy of ssDNA from FECs at different temperatures (more details in the SI). The conformational entropy initially increases, as shown in Figure 7B. However, further increasing c_{salt} leads to the charge inversion of ssDNA, which increases the extension by strengthening the electrostatic screening. As shown in Figure 7B, the conformational entropy decreases decreased after c^*_{salt}.

The Role of Hydrogen Bonds in ssDNA Charge Inversion. The RDF of oxygen atoms of water molecules with respect to every ion of the system was analyzed to characterize the hydration of ions in different salt concentrations (Figure S16). At intermediate salt concentrations, monovalent cations remain largely hydrated, and water molecules can bridge between the phosphate oxygens of DNA and the cations in their hydration shells, maintaining an intact hydrogen-bond network between DNA and water. At high salt concentrations, however, partial dehydration occurs due to excluded-volume effects, weakening this network as cations localize near the DNA surface. Such localization disrupts hydrogen bonds between DNA and the surrounding water molecules, ultimately leading to overcompensation of the backbone charge. The decrease in the number of hydrogen bonds between DNA and water molecules from our all-atom MD simulations confirms that increasing salt concentration promotes cation accumulation and progressively reduces DNA-water hydrogen bonding (Figure S17).

■ DISCUSSION

Our study uncovers a robust and previously unrecognized phenomenon: ssDNA undergoes charge inversion in the presence of high concentrations of monovalent salts, a regime overlooked by classical models of nucleic acid electrostatics. Using three orthogonal experimental approaches, including nanopore electrophoresis, DLS, and single-molecule MT experiments, we demonstrated that beyond a critical threshold concentration (~3 M), ssDNA reverses its net charge due to counterion overcompensation. This reversal leads to the reemergence of electrostatic repulsion, manifested as increased extension and persistence length. This phenomenon was consistently observed across different ssDNA lengths, sequences, and monovalent salts, including LiCl, NaCl, RbCl, and CsCl. While KCl did not induce charge inversion within experimentally accessible concentrations due to its low solubility, high-concentration KAc recovered the charge inversion, reinforcing the generality of this phenomenon. The all-atom MD simulations further supported our findings by reproducing the experimentally observed biphasic elastic response and quantifying both the magnitude and electrostatics of charge inversion.

While charge inversion has been reported for double-stranded DNA (dsDNA) in the presence of multivalent salts,[41] it was not previously believed to occur for ssDNA with monovalent salts. Our findings challenge the long-standing assumptions in nucleic acid biophysics. Classical theories, including those proposed by Murphy et al., Chen et al, and Ritort et al., reported a monotonic decrease in ssDNA persistence length with increasing monovalent salt concentration up to ~3 M.[13,14,21,22] However, this study reveals that this trend reverses at higher concentrations, a regime previously unexplored. Similarly, coarse-grained simulations by Wang et al. and mechanical studies by Saleh et al. focused on ionic strengths below the charge inversion threshold, thereby missing the charge inversion behavior entirely.[8,16,24,50] Our work significantly extends this boundary, showing that at sufficiently high monovalent salt concentrations, ssDNA charge inversion reintroduces electrostatic repulsions and fundamentally alters ssDNA conformation.

Our high-resolution electrophoretic and MT experiments clearly show that monovalent salts alone are sufficient to drive ssDNA into the charge inversion regime, in stark contrast to predictions from Debye−Hückel-theory-based expectations. Theoretically, ssDNA persistence length (L_p) is decomposed into the intrinsic and electrostatic components, with the bare, non-electrostatic persistence length ($L_0 \approx 0.6$ nm) and the electrostatic persistence length (L_e) capturing the electrostatic contribution between like charges.[16,51] The existing models, like the OSF model and the Barrat-Joanny (BJ) model, predicted that L_e scales inversely with salt concentration, $L_e \sim c_{salt}^{-1}$ or $L_e \sim c_{salt}^{-0.5}$, respectively.[43,44,52] These relationships are derived assuming monotonic exponential screening of electrostatics over the Debye length (λ_D), $\lambda_D \sim c_{salt}^{-0.5}$. The observation of nonmonotonic elasticity and charge inversion beyond ~3 M exposes fundamental limitations of such mean-field approaches. Our temperature-dependent experiments reveal that the conformational entropy of ssDNA increases with rising salt concentration. However, as the salt concentration exceeds a certain threshold and charge inversion occurs, the conformational entropy begins to decrease. These findings provide a more refined understanding of ssDNA behavior. Our results suggest that ion−ion correlations and strong ion-polymer interactions, which are neglected in traditional theories, must be incorporated into the model describing ssDNA electrostatics under extreme conditions.

Although the salt concentrations studied in this work are much higher than those under physiological conditions, such concentrations are achievable and used in a variety of applications in nanobiotechnology and materials processing. One application example is the nanopore translocation experiments, which rely on the changes in current. Higher LiCl concentration can enlarge the baseline current signal and thus improve the signal sensitivity, making it easier to detect the changes.[39] In addition, saturated ammonium sulfate solutions (~4.1 M) are commonly used for protein and protein−nucleic acid complex precipitation and crystallization, while multilayer DNA origami structures have been successfully assembled at NaCl concentrations as high as 2.4 M.[53,54]

Importantly, this work introduces a new electrostatic regime for ssDNA, which provides alternative approaches to investigate the effects of salt conditions on DNA nanostructures. ssDNA is present as the starting point of self-assembly for three-dimensional (3D) nanostructures, staples of DNA origami, or building blocks of two-dimensional (2D) tiles or 3D bricks.[55] As the precise control of DNA nanostructures is central to nanotechnology, the strategy of controlling the length and shape of ssDNA using salt is modular and simpler.[56−59] Typically, the DNA brick is composed of a 32/42-nt ssDNA with four consecutive domains that hybridize to complementary domains of other strands. These strands constitute the nanostructures with a periodicity of 6H (helix) by 6H (helix) by 48B (bp) cuboid.[54] The design of DNA bricks is extremely flexible, allowing assembly into any 3D

https://doi.org/10.1021/acs.macromol.5c02249
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structure. One can construct distinct shapes by adjusting the salt environment. In addition, the sub-1 nm technology node has utilized self-assembled 3D DNA patterns and DNA-directed nanofabrication, which requires the line pitch to be scaled down to sub-10 nm and the assembly to be correct and defect-free.[60,61] These crystallographic defects are generally introduced by the intrinsic physical properties or local swinging of individual DNA building blocks and are obvious within hundreds of misassembled DNA structures.[62] This line defect rate can be largely decreased with a larger periodicity of length, which should be accomplished by the use of salt conditions for ssDNA. Furthermore, the self-assembly of DNA bricks is limited by the nucleation step, where the nucleation barrier controls the rational kinetics of assembly.[63,64] The formation of random nuclei is a rare event and causes slow assembly in a bulk solution. The change in salt environment could reduce the electrostatic repulsions and favor the initial nucleation, or improve the stability of the preintroduced DNA "seed" strand that triggers fast nucleation.

Finally, ssDNA charge inversion can tune the assembly pathways and optimize the formation of specific structural motifs for DNA origami and other nanostructures. The spacing of ssDNA in condensates or scaffolded DNA structures could be manipulated by carefully controlling salt concentrations that promote or inhibit hybridization between specific strands, enabling the design of DNA devices that can switch between different conformational states in response to environmental triggers. These DNA nanostructures have the potential to create more responsive and adaptable DNA systems with applications in molecular machines, smart materials, and responsive drug delivery vehicles.[55−58] These findings should pave the way for the development of DNA-based materials with tunable flexibility, strength, and stability.

Charge inversion is a general result of strong electrostatic correlations and counterion condensation in concentrated electrolytes.[26,49] Therefore, our findings provide mechanistic insights for synthetic polyelectrolytes under similar high-salt conditions (e.g., strongly charged sulfonated or carboxylated polymers). Once the available salt concentration exceeds the correlation-dominated regime, synthetic analogs may exhibit mobility sign reversal and nonmonotonic elasticity. The onset and magnitude of the inversions are expected to depend on (i) linear charge density/Manning parameter and backbone chemistry, (ii) ion specificity and hydration (including partial counterion dehydration), and (iii) structure and local crowding effects.[25,45] Our findings offer new approaches to program chain stiffness, electrophoretic mobility, and interchain interactions in synthetic polyelectrolytes, enabling tunable assembly and transport in high-salt environments.

In conclusion, this study demonstrates that high concentrations of monovalent salts can induce ssDNA charge inversion, fundamentally altering its electrostatic and mechanical behavior. This finding expands the current understanding of nucleic acid physics and opens new directions for the controlled manipulation of DNA in both natural and engineered high-salt environments.

ASSOCIATED CONTENT

Data Availability Statement

All data needed to evaluate the conclusions in the paper are present in the paper and the Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.macromol.5c02249.

> Protocol of double-covalently linked ssDNA for MT experiments; methods for calculating the fraction of unpaired bases and entropy change versus temperature and force; derivation of the logarithmic equation of DNA charge; Figures S1−S17 (preparation schematic; force−extension curves across temperatures, monovalent salts, and divalent salts; sequence dependence; WLC fits; contour length per nucleotide; FEC integral; MD- and oxDNA-based analyses); additional references (PDF)

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Author Contributions

#W.W., F.-J.T. and Z.-X.C. contributed equally to this work. X.-H.Z. and L.D. conceived the idea. W.W. and Z.-X.C performed the experiments. L.D. and F.-J.T. performed the simulations. All authors discussed the results and wrote the paper.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We are grateful for the financial support from the National Natural Science Foundation of China (No. 12374216 and 12074294 to X.-H.Z.), the Hubei Provincial Natural Science Foundation of China (No. 2024AFE008 and 2025AFA021 to X.-H.Z.), and the Research Grants Council of Hong Kong (No. 11313322, 1130724, PDFS2425-1S09).

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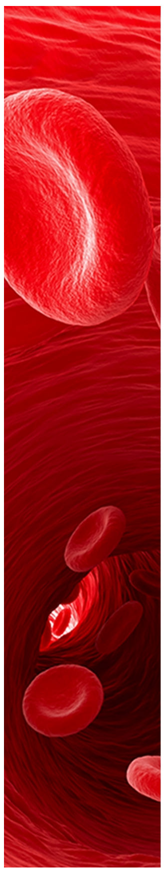

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https://doi.org/10.1021/acs.macromol.5c02249
Macromolecules 2025, 58, 12882−12892